

MOLTEN SALT REACTOR MICROSCOPIC DELAYED NEUTRON PRECURSOR GROUP
FORMULATION AND EVALUATION

BY

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DISSERTATION

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Abstract

fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs are delayed neutron precursors (DNPs), which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. DNPs are important for modeling and simulation of MSRs.

Many studies have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. These studies do not account for the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. The effect of neglecting these MSR-specific phenomena is reduced accuracy in the DNP group parameters.

These phenomena are important to consider, as simulations use the DNP group parameters to enable safety, security, and safeguards by design. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for developing an advanced reactor pipeline.

The DNP group parameters for MSRs are resolved using the microscopic, or aggregate or summation, method. This method entails using individual DNP data to reconstruct the DNP groups by summing them together in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half life and are used to generate the longer lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs have a non-negligible impact. Therefore, the longer lived DNP group parameters will vary, while the shorter lived DNP group parameters will remain the same.

A non-linear least-squares solver is used to generate a χ^2 minimized set of DNP group parameters. A specific DNP group parameter set is proposed for a particular set of conditions in the Molten Salt Reactor Experiment (MSRE). Verification is conducted via comparison with total delayed neutron yields and average half-lives from the literature. Validation is conducted via comparison with the MSRE during pump start-up and coast-down.

The results demonstrate the validity of these MSR phenomena-inclusive DNP group parameters. The methodology used to generate these parameters can be replicated for other MSR conditions, enabling the formation of more accurate sets of DNP group parameters for MSRs. This improves the accuracy of MSR simulations, which is important for safety, security, and safeguards by design as well as for developing an advanced reactor pipeline. Proposed work includes the following: development and evaluation of a more sophisticated model for generating the DNP group parameters, generation of polynomial fits for various parameters used in the DNP group generation, generation of DNP group spectra, uncertainty quantification, analysis of the flux shape effect on DNP group parameters, and evaluation of various datasets on DNP group parameters.

Acknowledgments

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Chapter 1

Introduction

- Go through history of group parameters (maybe timeline figure?).
- Discuss MSRs.
- "The next step forward for group parameter development is to have group parameters that accurately reflect the physics that occur in MSRs".
- Introduce the gap [148, 151, 78, 130, 72] that other works do not account for (they investigate movement of DNPs during transient, not for DNP group generation).
- Discuss methodology at a high level.
- Discuss results at a high level.
- Discuss conclusions of the work at a high level.

Overview of rest of document

1939 Roberts et al. discovery of delayed neutrons [114]. 1946 initial spectral measurements [24]. In 1947, Hughes et al. initial use of six groups (kinda) at Argonne heavy water pile [59]. In 1957, Keepin used a sample of U235 in Godiva to generate DNP group parameters (improved group fit). Thus far, only macroscopic approaches have been possible due to insufficient data. For MSRs which have chemical removal of certain elements, tracking individual nuclides is important. In 1989, Brady's thesis used updated data on 271 nuclides identified as DNPs (first attempt at microscopic/summation/aggregate approach). This data was used for ENDF-B/VI was released, and since then there have not been any changes to the DNP group parameters [143, 23]. In 1999, Parish says "Based on the six-group data reported in the literature, it appears that the ENDF/B-VI six-group constants are in need of revision." [99]. In 2014, International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [92]. 2020 Leconte uses more than 400 DNPs with data [77].

Brady's thesis has a good history (up to 1988).

This document should contain tabulated data for various fissile nuclides, neutron irradiation energies, in-core residence times, ex-core residence times, and mixing rates. Additionally, data should be provided on the exact concentrations of each nuclide (up to 99% of the DN yield).

Could also discuss various datasets. [87] - shows various datasets, most important nuclides, and the error associated with them (2012). [62] - Lots of delnu data (2014).

Chapter 2

Background and Motivation

This work aims to generate more accurate DNP group parameters for MSRs and implement those parameters for safety, security, and safeguards design purposes. These DNP group parameters in MSRs differ because fuel circulates and undergoes online chemical reprocessing. These effects can be captured in spatially resolved MSR fuel depletion simulations. The existing methodology for DNP group parameter generation does not account for the physical phenomena present in MSR fuel depletion [66, 135, 86, 13, 142, 21, 59, 84, 85, 106, 32, 106, 121, 143, 20, 77]. More accurate DNP group parameters are generated by accounting for these physical phenomena, which is particularly important for safety, security, and safeguards related concerns.

2.1 Delayed Neutron Precursors

Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually. DNPs are an important consideration in nuclear reactors [33]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [10]. In this work, the term DNP will be solely used rather than referring to emitters, as the delay of the neutron emission is effectively only due to the β decay [18, 98]. Classifying between the emitter and the DNP is not relevant for this work.

This delay in neutron emission significantly affects the reactor period and transient behavior. Equation (2.1) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.1)$$

where,

$\dot{n}_{d,i}$ = the delayed neutron activity, or count rate, from DNP i [$\frac{n}{s}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from
nuclide i , or the emission probability [-]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of DNP i [$atoms$].

In MSRs, the movement of the fuel salt necessitates spatial resolution for depletion, transmutation, and neutron emission. In a solid fuel reactor, such as a light water reactor (LWR), a spatial component is not necessary to properly model the DNPs. Modeling space and time together causes computational costs to increase significantly, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity worth, and reactor controllability [33, 34, 73]. Section 2.3.1 shows how the power response and reactivity worth depend on the DNPs. The reactor controllability varies based on the available number of DNPs. If there are fewer delayed neutrons available due to movement of the fuel salt, such as in a MSR, then the reactor period becomes shorter and makes the reactor harder to control.

2.1.1 Measurement

Macroscopic and microscopic approaches are used for measuring DNPs [99, 121]. The macroscopic approach uses a fissile sample irradiation to get the delayed neutron count rate, which can be approximated using groups. The microscopic approach instead uses the data from many experiments combined together to simulate a fissile sample irradiation and generate the delayed neutron count rate. This makes the microscopic approach more challenging, as it depends on the work of many experimentalists, though it also offers more detail on the delayed neutron count rate.

Macroscopic Measurement of Delayed Neutron Precursors

This section covers the macroscopic approach. Analysis and collection of macroscopic experiment data is very common in the literature [66, 135, 86, 13, 142, 21, 59, 84, 85, 30]. The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. Small samples are used in order to minimize neutron multiplication within the sample. Two different types of irradiation are used: pulse irradiation, where the irradiation time is shorter than the shortest half-life; and infinite irradiation, where the irradiation time is longer than the longest half-life. After irradiation, the delayed neutrons from the sample are counted as a function of time. The delayed neutron count rate as a function of time after irradiation, $\dot{n}_d(t)$, is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where,

$$\dot{n}_d(t) = \text{the delayed neutron count rate} \left[\frac{n}{s} \right]$$

$$I = \text{the total number of DNPs [-]}$$

$$\epsilon = \text{the detector efficiency [-]}.$$

Once this delayed neutron count rate is generated, it is common to then use the data to generate group and kinetics parameters for that fissile nuclide and neutron energy. The specifics of the methods used to form those parameters is discussed in Section 2.1.2. Figure 2.1 (recreated from [66]) shows an experimental setup for collecting delayed neutrons.

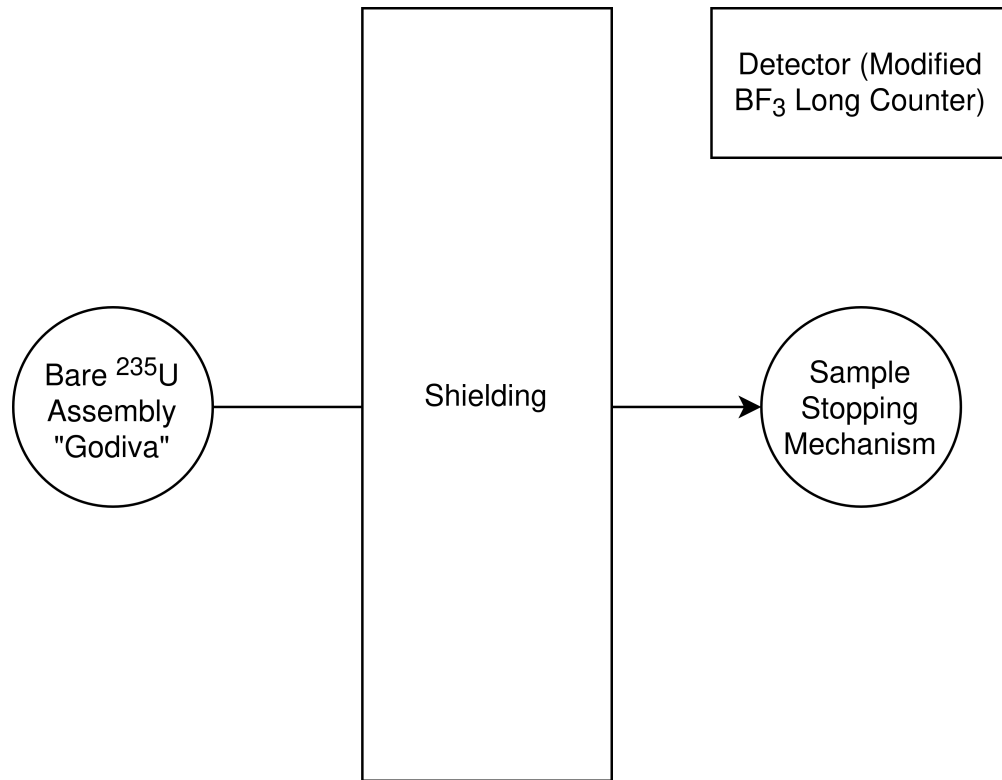


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [66].

Challenges associated collecting this experimental data [10] include experimental noise and signal obfuscation due to moderation. Experimental noise becomes significant after long periods of time, as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation. This means that both the long and short time-scale measurements have greater uncertainty than the middle time-scale measurements.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, this does not account for source multiplication within the irradiated sample. Keepin et al. suggests a technique to correct for the sample self-multiplication. The sample self-multiplication require a correction factor of up to 3% in the number of fissions.

Microscopic Measurement of Delayed Neutron Precursors

The other approach used to measure DNPs is the microscopic approach, investigated in detail in the 1990-2000's [106, 32, 121, 143, 20], with progress through the 2010's with more data [92, 62]. Leconte et al. used this approach as recently as 2020 [77]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape in order to rapidly measure short lived nuclides. The magnetic tape is used to transport samples or remove longer-lived decay products [127, 122].

There are many different experimental methods used to study DNPs, particularly their spectra: on-line mass spectrometry, helium-3 spectrometry, the proton-recoil technique, the time-of-flight method, and the neutron

scintillator [33]. On-line mass spectrometry combined with a helium-3 ionization chamber and the proton-recoil technique allows for measurement of the delayed neutron energy spectrum [126, 16]. The time-of-flight method involves measuring the velocity of the delayed neutron by having a known travel distance and time measurement [29, 47, 28]. An example of the time-of-flight experimental setup is shown in Figure 2.2.

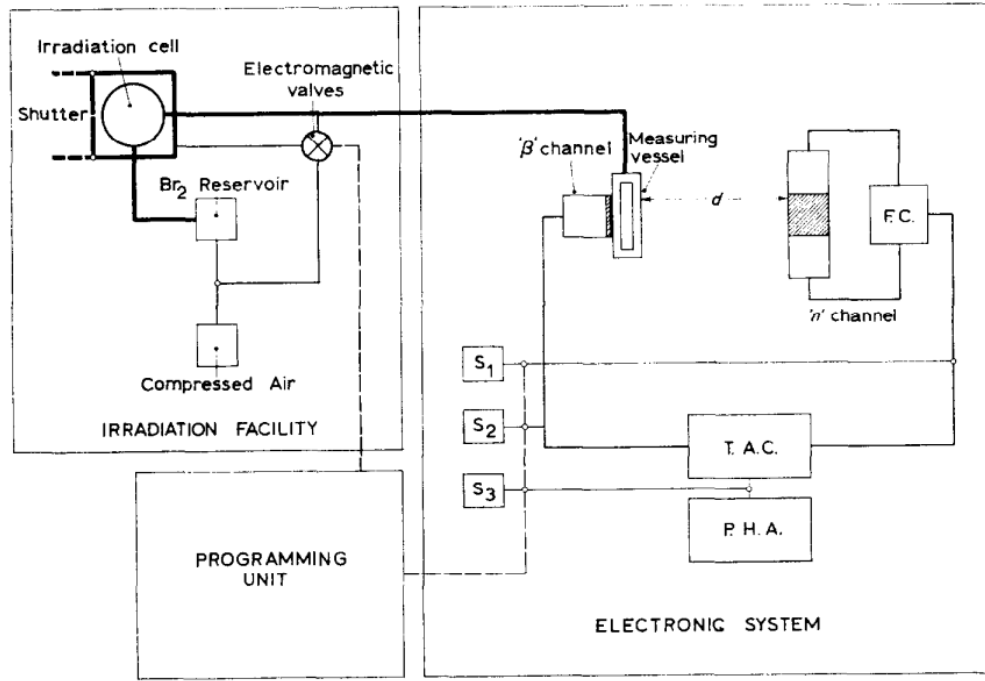


Figure 2.2: A block diagram of the time-of-flight experimental setup for measuring delayed neutrons from bromine-87 and bromine-88, extracted from the Democritos reactor via a fast radiochemical technique recreated from [28].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. Fortunately, the IAEA has worked to compile and evaluate this data into a publicly accessible form [37, 36]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly, rather than using groups. This is particularly useful for understanding what nuclides have the greatest impact as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

2.1.2 Delayed Neutron Precursor Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally understanding delayed neutron precursors were discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The common way of representing the DNP groups is in several distinct groups, typically six. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and England found that increasing the number of groups for point kinetics calculations from six to nine caused no significant differences [20]. This shows that merely increasing the number of DNP groups does not provide an increase in accuracy for point kinetics simulations.

This section goes through the calculation of DNP group parameters and kinetics parameters. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.1 goes through each of the terms of importance for this section.

Parameter	Description
λ_k / λ_i	The decay constant for DNP group k / nuclide i
$\bar{\nu}_{d,k} / \bar{\nu}_{d,i}$	The delayed neutron yield for DNP group k / nuclide i
χ_k / χ_i	The spectra of DNP group k / nuclide i
$P_{n,i}$	The delayed neutron emission probability for nuclide i
$\langle T \rangle$	The mean half-life
Y	The total delayed neutron yield
β_k / β_i	The delayed neutron fraction of DNP group k / nuclide i
β	The total delayed neutron fraction
β_{eff}	The effective delayed neutron fraction
Λ	The mean neutron generation time
Λ_{eff}	The effective mean neutron generation time

Table 2.1: DNP group parameters and kinetics parameters along with their definitions.

The DNP groups come with associated parameters, such as the group decay constants, λ_k , and the group yields, $\bar{\nu}_{d,k}$. These group yields can be converted to group delayed neutron fractions, β_k , by dividing the group yields by the average number of neutrons emitted per fission event. The group abundances represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_k and λ_k terms are the defining parameters for the DNP groups, along with the group spectra, χ_k , for DNP group k . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs nuclides are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [104]. In 1989, Brady and England also use the individual nuclides to compare with group data [20], and show that the updated six-group parameters offer consistent results with the explicit DNPs. Updating the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. Arguably the most well known term in nuclear engineering is the effective multiplication factor, k_{eff} . This term represents the scaling of the number of neutrons over each generation. There is also the infinite multiplication factor, k_∞ , which represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction and mean generation time terms, where the effective form of the term requires information on the specific reactor geometry.

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move away from the center of the core due to fuel salt advection.

The delayed neutron fraction, β , represents the average number of delayed neutrons emitted per fission event, or the ratio of delayed neutron production over all neutron production [91]. Related to the delayed neutron fraction

is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events [119]. Alternatively, it can be thought of as the ratio of the spectrum and adjoint weighted delayed neutron production over all spectrum and adjoint weighted neutron production [91]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

Another term, the mean neutron generation time, Λ , is the average "...time between the birth of a neutron and the subsequent absorption that induces fission" [119]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [137]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [137].

2.1.3 DNP Group Parameters

The DNP group parameters are important for simulating transients. The DNP group parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.2) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for I individual DNPs for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t)$$

Equation (2.3) instead gives the K group DNP fit measured by the same detector for a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.3)$$

For an infinite, or saturation, irradiation, Equation (2.4) is used rather than Equation (2.3), given as [21]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t}. \quad (2.4)$$

The terms in Equations (2.3) and (2.4) are defined as follows:

- K = the number of DNP groups [-]
- λ_k = the decay constant of DNP group k [s^{-1}]
- F_s = the total number of fission events in the pulse irradiation [-]
- $\dot{F}_s$ = the fission rate for a saturation irradiation [s^{-1}]
- $\bar{v}_{d,k}$ = the group yield, or group abundance for group k [-] .

More information on the derivation of these equations can be found in Appendix A. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [143]. To make this

equation generic, the $\bar{v}_{d,k}\lambda_k$ term and $\bar{v}_{d,k}$ term can instead be written as A_k . Additionally, the fission term can be written as $\bar{F}_s$, to represent either a fission rate or a net number of fissions. These equations are simplifications, as they do not account for decay chain daughter products [30].

The pulse irradiation approach gives better statistics on the short lives DNPs, while the longer lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [21]. Specifically, Brady uses the four long-period groups from the saturation irradiation and the four short-period groups from the pulse irradiation renormalized at the second DNP group.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group abundances and decays. For example, Brady used STEPIT to handle the non-linear least-squares solve with both pulse and infinite irradiation experimental data [21]. Waldo et al. also solve the least-squares problem, though the specific approach used is not specified [142]. The general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.5)$$

Equation (2.6) gives the χ^2 term the non-linear least squares minimizes as:

$$\chi^2 = \sum_{m=0}^M \left(\frac{\vec{b}^{(m)} - (A\vec{x})^{(m)}}{\vec{\sigma}^{(m)}} \right)^2, \quad (2.6)$$

where,

χ^2 = the chi-squared measure of the fit

M = the total number of time steps over which the sample decays

σ = the uncertainty at time step m , or unity if the uncertainty is not measured.

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.3) and (2.4). Equation (2.7) gives the coefficient matrix for a pulse irradiation as:

$$A = \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{bmatrix}, \quad (2.7)$$

though for a saturation irradiation the only change would be removing the λ_i term.

Within this least squares equation, the solution vector contains only the group yields, given in Equation (2.8) as:

$$\vec{x} = \begin{bmatrix} \bar{v}_{d,1} \\ \bar{v}_{d,2} \\ \vdots \\ \bar{v}_{d,K} \end{bmatrix}. \quad (2.8)$$

The target vector contains the linearized delayed neutron count rate data normalized to the number of fissions, or fission rate, and the detector efficiency, as shown in Equation (2.9) as:

$$\vec{b} = \begin{bmatrix} \ln(n_d(t_0)(\epsilon F_s)^{-1}) \\ \ln(n_d(t_1)(\epsilon F_s)^{-1}) \\ \vdots \\ \ln(n_d(t_m)(\epsilon F_s)^{-1}) \end{bmatrix}. \quad (2.9)$$

By solving this non-linear system of equations, the group half-lives, T_n , and abundances, a_n , can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, F , then Equation (2.10) can be used to determine the K group half lives using [33] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.10)$$

where,

$$\begin{aligned} \lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F. \end{aligned}$$

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Equation (2.11) shows that including the energy in the delayed neutron count rate requires the energy spectra, $\chi_{d,k}(E)$, of the relevant group to be included. This is also the general form of the saturation and pulse irradiation equations, as it includes A_k , which can represent $\bar{v}_{d,k}\lambda_k$ or $\bar{v}_{d,k}$ depending on the irradiation type. $\tilde{F}_s$ is the total number of fissions for a pulse irradiation, or the fission rate, for saturation irradiation. The energy and time dependent neutron count rate is given as:

$$\dot{n}_d(E, t) \approx \epsilon \tilde{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.11)$$

where,

$$\begin{aligned} \tilde{F}_s &= \text{the generic fission term [- or } s^{-1} \text{]} \\ A_k &= \text{the generic matrix term [} s^{-1} \text{ or -]} \\ \chi_{d,k}(E) &= \text{the delayed neutron energy spectrum of DNP group } k \text{ [-]}. \end{aligned}$$

2.1.4 DNP Group Spectra and Concentrations

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is important for using the generated group parameters with concentrations which are converted into group concentrations. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

Two different methods are considered for separating and combining the DNP groups into individual nuclides and vice versa. These methods have been used to generate DNP group spectra, so the group spectra can also be calculated using these methods:

- Grouping based on half lives [40, 120]

- Grouping based on fractional contribution to adjacent groups [21]

The grouping based on half lives requires comparing each nuclide's half life to a constant value. The groupings based on half lives are given in the top two sets in Table 2.2. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach.

Table 2.2: Half-lives of DNP groups from various sources.

Source	T_1 [s]	T_2 [s]	T_3 [s]	T_4 [s]	T_5 [s]	T_6 [s]
[40]	$T > 31$	$31 \geq T > 9$	$9 \geq T > 3$	$3 \geq T > 0.8$	$0.8 \geq T > 0.3$	$0.3 \geq T$
[120]*	$T > 30$	$30 \geq T > 10$	$10 \geq T > 4$	$4 \geq T > 1.4$	$1.4 \geq T > 0.4$	$0.4 \geq T$
[23]	$T = 51.98$	$T = 21.17$	$T = 5.74$	$T = 2.29$	$T = 0.82$	$T = 0.24$

* Rudstam uses $T_1 = 55.6$, but for the sake of completeness, this has been modified.

The half-life grouping method thus results in a series of comparisons. If a nuclide is within a given group's bounds, the concentration of that nuclide is then added to that group.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups, i and $i + 1$, for each nuclide k . These groups are determined via Equation (2.12) as:

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.12)$$

Nuclides which don't fit in these bounds will have the entire fractional contribution go the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.13) as:

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.13)$$

To solve for the fractions, the least squares minimization problem is solved as:

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.14)$$

Solving this equation for where the partial derivatives are zero for each fraction, $f_{k,i}$ and $f_{k,i+1}$, combined with Equation (2.13), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} \\ \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{2\lambda_i \lambda_k}{\lambda_i + \lambda_k} \\ \frac{2\lambda_k \lambda_{i+1}}{\lambda_k + \lambda_{i+1}} \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix}. \quad (2.15)$$

Because the number of rows is greater than the number of columns in the matrix, the left-inverse is used. This system of equations is then solved as shown in Equation (2.16) as:

$$x = (A^T A)^{-1} A^T b \quad (2.16)$$

With the fractional contribution terms, x , the concentration of a given nuclide can then be multiplied by each fraction to show much it contributes to that group. With a summation of each nuclides contribution, the group concentrations can thus be calculated and vice versa. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

2.1.5 Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The k^{th} DNP group delayed neutron fraction is calculated as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}. \quad (2.17)$$

A useful term for comparing DNP group parameters is the average half-life [107, 108]. Equation (2.18) gives the average half-life as:

$$\langle T \rangle = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_i}. \quad (2.18)$$

Equation (2.19) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event, $\bar{v}_d$, can be calculated [51]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.19)$$

This yield can also be approximately represented for an individual nuclide by integrating Equation (2.1) over all time, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0)e^{-\lambda_i t}. \quad (2.20)$$

This yields Equation (2.21), where the total number of delayed neutrons emitted from a single nuclide i is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.21)$$

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{v}_d}{\bar{v}} \quad (2.22)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.23)$$

where,

l = the prompt neutron lifetime [s]

k = the neutron multiplication factor [-].

(2.24)

This can be calculated using a time-dependent version of a Monte Carlo code along with point reactor kinetics (PRKs) [119].

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} can be calculated explicitly by its definition, which is "...the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events." This is given in Equation (2.25) as [119]:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}, \quad (2.25)$$

where,

$$\begin{aligned} \Phi^\dagger &= \text{the adjoint neutron flux } [\frac{n}{cm^2s}] \\ \chi_d &= \text{the delayed neutron energy spectrum } [-] \\ \chi &= \text{the neutron energy spectrum } [-] \\ \beta &= \text{the delayed neutron fraction } [-] \\ \bar{\nu} &= \text{the average number of neutrons released per fission } [-] \\ \Sigma_f &= \text{the macroscopic fission cross section } [cm^{-1}] \\ \Phi &= \text{the neutron flux } [\frac{n}{cm^2s}]. \end{aligned}$$

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.26) as [137]:

$$\Lambda_{eff} = \frac{\int \Phi_0^\dagger \frac{1}{\bar{\nu}} \Phi_0 dE d\vec{\Omega} d\vec{r}}{\int \Phi_0^\dagger F_0 \Phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.26)$$

where,

$$\begin{aligned} \Phi_0^\dagger &= \text{the critical adjoint neutron flux } [\frac{n}{cm^2s}] \\ \Phi_0 &= \text{the critical neutron flux } [\frac{n}{cm^2s}] \\ \bar{\nu} &= \text{the average neutron speed } [\frac{cm}{s}] \\ F_0 &= \text{the total fission operator } [cm^{-1}]. \end{aligned}$$

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [137]. These include the direct integration, or adjoint weighted, method [67]; the eigenvalue, or perturbation, method [56]; and the iterated fission probability method [17]. There is also the pulsed method for calculating Λ_{eff} , and the prompt method for calculating β_{eff} [119]. For β_{eff} , there are also the Meulekamp and Spriggs methods which can be used [91]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [8].

2.1.6 Kinetics and Dynamics Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [144]. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena when generating the DNP group parameters.

Multi-group diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches which can be applied to transient problems, surpassed only by transport. Hurwitz and Ehrlich provide a standard formulation of multi-group diffusion written in 1953 [61].

Examples of works that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [44, 132]. Other examples can be found in the review by Wooten et al. [70, 45, 69]. Even more works which use the multi-group diffusion method can be found in the literature [148, 146, 71, 95, 101].

These works typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling is given as [144]:

$$\begin{aligned} \frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_{g'}, \end{aligned} \quad (2.27)$$

and,

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\bar{v}_g \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i, \quad (2.28)$$

where,

- $\bar{v}_g$ = the average number of neutrons produced by fission of group g [-]
- ϕ_g = the scalar neutron flux of group g [$\frac{n}{cm^2s}$]
- D_g = the group g diffusion term [cm]
- G = the total number of energy groups [-]
- $\Sigma_{s,g' \rightarrow g}$ = the macroscopic scattering cross section from group g' to group g [cm^{-1}]
- $\chi_{p,g}$ = the number of prompt neutrons born in group g [-]
- $\chi_{d,g,i}$ = the number of delayed neutrons from precursor group i born in group g [-]
- C_i = the concentration of DNPs in group i [$\frac{atoms}{cm^3}$]
- $\vec{u}$ = the velocity profile [$\frac{cm}{s}$]
- $\Sigma_{a,g}$ = the macroscopic absorption cross section for group g [cm^{-1}]
- ν_T = the eddy viscosity [$\frac{cm^2}{s}$]
- Sc_T = the turbulent Schmidt number [-].

Modified point kinetics

PRK are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t), \quad (2.29)$$

and,

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t), \quad (2.30)$$

where,

- n = the relative neutron density [-]
 - ρ = the reactivity [-]
 - β_{eff} = the effective delayed neutron fraction [-].
- (2.31)

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [56, 74]. Modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [144]. Wooten et al. provide four different works which use MPK in MSR analysis [52, 90, 129, 147].

The work by Guo et al. analyzes unprotected loss of heat sink (ULOHS) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics (DPK), which model precursor concentrations in-core and ex-core, with the

primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,c}(t), \quad (2.32)$$

$$\begin{aligned} \frac{dC_{i,c}(t)}{dt} &= \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_c} \right) C_{i,c}(t) \\ &\quad + \frac{1}{\tau_l} \left(\frac{V_l}{V_c} \right) C_{i,l}(t - \tau_l), \end{aligned} \quad (2.33)$$

$$\frac{dC_{i,l}(t)}{dt} = \frac{1}{\tau_c} \left(\frac{V_c}{V_l} \right) C_{i,c}(t) - \left(\lambda_i + \frac{1}{\tau_l} \right) C_{i,l}(t). \quad (2.34)$$

These equations, shown in Equations (2.32) through (2.34) (recreated from [52, 129]), are also used by Shi et al. in the Reactor Excursion and Leak Analysis Program (RELAP) for a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [129]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but fails to capture all oscillatory effects [144].

Intermediate terms used in Equations (2.32) through (2.34) are shown in Equations (2.35) through (2.38) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \quad (2.35)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,l}, \quad (2.36)$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \quad (2.37)$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}. \quad (2.38)$$

The terms in Equations (2.32) through (2.38) are defined as follows:

V_l = the volume of the loop [cm^3]

V_c = the volume of the core [cm^3]

β_{static} = the value of β_{eff} without flow [-]

M_c = the mass in the core [kg]

M_l = the mass in the loop [kg]

$\dot{M}(t)$ = the mass flow rate [$\frac{kg}{s}$].

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [131]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses a method called "IPK method," which may stand for "Improve Point-Kinetic method," though this is not explicitly stated [90]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [90].

The work by Zanetti et al. extends the TRACE code, a finite volume code used for LWR transients [147]. The

DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [75]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich, and used in PRK equations [55].

Quasi-statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [144]. The general formulation used for quasi-static methods is very similar among the various works that use it [38, 132]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [56, 35]. There are also other works which discuss the quasi-static method in detail [12].

Wooten et al. introduce several different works which have used quasi-statics [144]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [113]. The neutron flux is constrained in the work by a term containing the phase-space integration and the weighting function used. The neutron flux is treated as:

$$\Phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.39)$$

where,

$$\begin{aligned} \Phi &= \text{the neutron flux } [\frac{n}{cm^2s}] \\ \phi_A &= \text{the neutron flux amplitude function [n]} \\ \psi(\vec{x}, t) &= \text{the neutron flux shape function } [cm^{-2}s^{-1}]. \end{aligned}$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.40) and (2.41). These equations are solved by time-discretizing Equation (2.40) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5]. These primary equations are given as:

$$\begin{aligned} \frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v}\phi_A} \frac{d\phi_A}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta) \chi_p F(t) \psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t), \end{aligned} \quad (2.40)$$

and,

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = \phi_A \beta_i F \psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t) C_i(\vec{x}, t)). \quad (2.41)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters are calculated as well as the flux amplitude. The equations used in the "reactivity" steps are

recreated here as:

$$\frac{d\phi_A}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}), \quad (2.42)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_i c_{s,i}. \quad (2.43)$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.44)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d c_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.45)$$

Intermediate terms used in Equations (2.42) through (2.45) are given as:

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla (u(\vec{x}, t) C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla (u(\vec{x}, t) C_{m,i}(\vec{x}, t)), \quad (2.46)$$

and,

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i C_{s,i}(\vec{x}, t). \quad (2.47)$$

The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.40) and (2.41) to zero as well as the fluid velocity to zero at $t=0$. This allows Equation (2.46) to be solved.

The terms in Equations (2.39) through (2.47) are defined as follows:

- $M(t)$ = the neutron disappearance operator [cm^{-1}]
- $F(t)$ = the neutron production operator [cm^{-1}]
- χ_p = the prompt neutron energy spectra [-]
- $C_i(\vec{x}, t)$ = the concentration of DNP group i [$\frac{atoms}{cm^3}$]
- $Q(\vec{x}, t)$ = the external neutron source [$\frac{n}{cm^3s}$]
- $u(\vec{x}, t)$ = the DNP velocity [$\frac{cm}{s}$]
- $C_{s,i}(\vec{x}, t)$ = the "standard" precursor concentration of group i [$\frac{atoms}{cm^3}$]
- $C_{m,i}(\vec{x}, t)$ = the "movable" precursor concentration of group i [$\frac{atoms}{cm^3}$]
- ρ = the reactivity [-]
- $c_{s,i}$ = the PRK "standard" precursor concentration of group i [$atoms$]
- $c_{m,i}$ = the PRK "movable" precursor concentration of group i [$atoms$]
- q_{eff} = the PRK external neutron source [$\frac{n}{s}$]
- Λ = the neutron generation time [s]
- $W(x)$ = the weighting function (the default in SIMMER
is an adjoint flux computed at steady-state conditions) [-].

2.2 Molten Salt Reactor Fuel Depletion

There are many different MSR designs in the literature, as shown in Table 2.3 [27]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, causes depletion to vary spatially far more than stationary fuel, movement of DNPs, and close coupling of thermal-

hydraulics to neutronics.

MSR Name	Source
FUJI series	[46]
Actinides Molten Salt TransmutER (AMSTER)	[138]
MOSART	[63]
Fast Spectrum Molten Salt Reactor (FS-MSR)	[57]
European MSFR	[89]
Small Mobile Molten Salt Reactor (SM-MSR)	[25]
MSBR	[115]
MSRE	[116]
Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor	[19]
Single fluid and two fluid denatured molten salt reactors (DMSRs)	[39, 4]
Thorium molten salt reactor (TMSR)	[97, 88]

Table 2.3: A collection of various MSR designs publicly available in the literature.

There are many works in the literature which have investigated online chemical reprocessing of MSRs, as can be seen in Table 2.4 recreated from [124]. Table 2.4 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, which shows most models do not include DNP movement in depletion. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models and is also sorted by the publication year in the "Year" column. The table shows that there are many MSR concepts, shown in the "Reactor" column, for which depletion was modeled. The study by Valdez et al. investigates two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.4: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [124]).

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [65]	Continuous	Unit Cell	Several	No	2020
Zhuang et al. [150]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [145]	Continuous	Full 3D	MSFR	No	2019
Zhou et al. [149]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [9]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [117]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [123]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [15]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [112]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [100]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [128]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [96]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [43]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [94]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [88]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [53]	Mixed	2D Slice	MSBR	Yes	1971

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

2.2.1 Depletion Simulations

Depletion is the process of fission, transmutation, decay, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [79]. Simulating depletion entails numerically solving the Bateman equations; a generalized form of which is given as [79]:

$$\begin{aligned} \frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.48)$$

where,

- N_j = the atomic density of isotope j $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i $[\text{cm}^2]$
- ϕ = the spectrum-averaged scalar flux in the fuel region $\left[\frac{\text{neutrons}}{\text{cm}^2 \text{s}} \right]$
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ $[\text{s}^{-1}]$
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[\text{cm}^2]$
- N_i = the atomic density of isotope i $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- λ_j = the decay constant of isotope j $[\text{s}^{-1}]$
- $\lambda_{r,j}$ = the removal constant for removal of isotope j $[\text{s}^{-1}]$
- σ_j = the microscopic total transmutation cross-section of isotope j $[\text{cm}^2]$
- $\dot{N}_{f,j}$ = the feed rate for isotope j $\left[\frac{\text{atoms}}{\text{cm}^3 \cdot \text{s}} \right]$.

The matrix form is given as:

$$\frac{d\vec{N}}{dt} = A\vec{N}, \quad (2.49)$$

where,

- $\vec{N}$ = the vector of material compositions $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- A = the depletion matrix $[\text{s}^{-1}]$.

The depletion matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well. Further documentation on these terms can be found in the OpenMC documentation, an open source Monte Carlo transport code which has built-in depletion solvers [118].

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [80]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [96] to implement 4th order Runge–Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [9].

Overall, there are many approaches to solving the depletion equation with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.48) is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten as:

$$\frac{dN_j}{dt} = S(t) - \lambda_{net,j} N_j(t), \quad (2.50)$$

where,

$$\begin{aligned}
S &= \text{the source term for nuclide } j, \\
&\quad \text{containing fission, decay, and transmutation sources } [\frac{atoms}{cm^3 s}] \\
\lambda_{net,j} &= \text{the loss term for nuclide } j, \\
&\quad \text{containing decay and transmutation losses } [s^{-1}].
\end{aligned}$$

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. This also means that the loss term, $\lambda_{net,j}$, does not vary with time in this equation.

This form of the Bateman equation does not have space or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel makes the problem more complex, as adding a spatial component means that instead of a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.51)$$

where,

$$D_{mix} = \text{the mass diffusivity } [\frac{cm^2}{s}].$$

This equation assumes spatial resolution is only applied to the concentration terms, which means the depletion matrix does not include spatial resolution. Thus, only the concentrations and material blocks provide spatial resolution to the problem. More spatial refinement can be provided to the neutronic parameters based on the specific approach used, such as a fully spatially resolved neutron flux.

As previously mentioned, the main challenge faced when solving this problem is the computational cost. However, there are still many works which have attempted to develop methods to get around this problem. One approach, the scaled flux method, is computationally inexpensive [14]. However, this approach does not model the in-core and ex-core regions in a physical manner. Instead, the scaled flux method scales the flux by the fraction of fuel salt in-core compared to ex-core. Another approach is to calculate the steady-state concentration, changing the system of ordinary differential equations to spatial rather than temporal relations [125].

Other approaches, such as Monte Carlo N-Particle (MCNP) and Advanced Dimensional Depletion for Engineering of Reactors (ADDER), or the method proposed by Tano et al. using Griffin and Pronghorn, try to model the flow more explicitly [6, 133]. This approach is more computationally expensive than the other approaches that only require solving a system of ordinary differential equations. However, the literature has shown that including advection in depletion directly affects the equilibrium concentration of ^{135}Xe by varying amounts [133, 42]. Tano et al. show that the ^{135}Xe in the MSFR decreases by approximately 80% when moving from no spatial resolution to a quasi-static method. Fei et al. show that the ^{135}Xe in the MSRE increases by approximately 63% when moving

from no spatial resolution to 0D fuel flow. This means that depletion simulations of MSRs need to include spatially resolved depletion to generate accurate results.

However, including fully resolved spatial resolution in depletion has been referred to as computationally "intractable" [14, 133]. This is because tracking the ~ 2000 nuclides over space and time requires new methods rather than those used for the time dependent system of ordinary differential equations. Methods for solving the system of partial differential equations require accurately capturing short time scale flow effects, which makes multi-year long depletion simulations exceedingly expensive. Numerical stability of the problem, when using an explicit-time solving scheme, also requires the Courant-Friedrichs-Lewy (CFL) condition be satisfied as:

$$\lambda_{CFL} \geq \frac{u\Delta t}{\Delta z}, \quad (2.52)$$

where,

Δt = the time step size [s]

Δz = the length between spatial nodes [cm].

This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

However, for a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the computational cost is orders of magnitude smaller. This means that having a high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment to generate updated DNP group parameters is feasible.

2.3 Safety, Security, and Safeguards

2.3.1 Safety

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there are not specific events listed for MSRs, guidance is provided on categories of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, events which are not independent of each other could potentially occur at the same time.

Safety-related transients for MSRs include ULOHS, ULOF, unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [81, 31]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, which causes a loss of cooling of the reactor. For the MSFR,

Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [22]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results.

2.3.2 Security

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [110]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [7, 48]. There are several different categories which are important to understanding security: the actor type, the actor capabilities, their objectives, and the strategies they use to achieve said objectives. In general, the main consideration for security is protection of the public by preventing theft of weapons-usable nuclear material (WUNM), theft of hazardous radioactive material, and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [110].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

There are five categories of actor capabilities: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect their strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant more tools accessed with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the

deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, the specific actor capabilities and strategies are not considered. Additionally, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. This work investigates the formulation of new DNP group parameters, which impact the reactor behavior during a transient. This means that there may be security issues that arise during a sabotage-initiated transient due to the recalculated kinetics and DNP group parameters.

2.3.3 Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards, referred to as proliferation resistance (PR) here, is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [110]. This definition agrees with the definition made at the IAEA international workshop in 2002 [64].

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [110]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [110].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of 2023 that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [58]. Based on the more recent Order, special nuclear material (SNM) is classified as shown in Table 2.5.

Table 2.5: The SNM classification table is recreated from [58].

Material Type	Accountable Quantity	Weight Field Used for Element	Weight Field Used for Isotope
^{235}U	1 g	total uranium	^{235}U
^{233}U	1 g	total uranium	^{233}U
^{242}Pu	1 g	total plutonium	^{242}Pu
$^{239-241}\text{Pu}$	1 g	total plutonium	$^{239}\text{Pu} + ^{241}\text{Pu}$
^{238}Pu	0.1 g	total plutonium	^{238}Pu
$^{241}\text{Am}^{**}$	1 g	total americium	^{241}Am
$^{243}\text{Am}^{**}$	1 g	total americium	^{243}Am
$^{237}\text{Np}^{**}$	1 g	total neptunium	-

^{**} Americium and neptunium are treated as SNM if they are separated.

Table 2.5 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials,

Use	Material	Significant Quantity [kg]
Direct	Plutonium	8
Direct	HEU	25
Direct	^{233}U	8
Indirect	^{233}U	75
Indirect	Natural Uranium	10,000
Indirect	Depleted Uranium	20,000

Table 2.6: Significant quantities of various materials are given along with their use category.

low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% ^{235}U or <10% ^{233}U .

The quantities provided in Table 2.5 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or $^{233}\text{U} \geq 6$ kg is considered Category I SNM. Alternatively, if the same material contains ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am , then an amount ≥ 20 kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" do not have a listed Category I quantity for material containing ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am .

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). SQs of the different material categories are given in Table 2.6 [110, 3, 11].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

Chapter 3

Methodology

3.1 Tools

To evaluate and generate the updated DNP group parameters, several external software tools are required. In-house Python scripts are used to collect and combine various data, OpenMC generates the simplified group parameters, and the MOOSE app Moltres is used to both generate higher fidelity and evaluate lower fidelity updated group parameters [136, 118, 83]. Further information on each tool is provided in its corresponding sub-section.

3.1.1 Python Scripts

The Python scripts used in the simplified DNP group generation make use of the SciPy, NumPy, Matplotlib, pandas, and Uncertainties packages [140, 54, 60, 134, 41]. These scripts set up the OpenMC sample irradiation simulation, collect data from the IAEA beta-delayed neutron emission database, combine the OpenMC concentrations with the collected half-life and emission probability data, and fit group parameters to the calculated delayed neutron count curve [37]. Section 3.2 provides more details on what these scripts do.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutron and photon transport code. For the purposes of this work, OpenMC offers many benefits. The Python Application Programming Interface (API) provides useful functionality which makes interfacing with the in-house Python scripts straightforward. Irradiation of the sample is also simple and can be easily configured, adjusting various parameters by altering the Python script inputs. OpenMC also offers useful functionality in depletion, where a zero-power depletion step is rapidly calculated since only depletion is tracked. The transfer-rates functionality in OpenMC can also be used during depletion, which enables simulation of chemical removal that would occur in an MSR.

For this work, the scripts use the ENDF/B-VIII.0 based depletion chains provided by OpenMC and ENDF/B-VIII.0 cross section data. The thermal spectrum chain is used for neutron irradiation less than 300 eV, while the fast spectrum is used for higher energy neutron irradiation. For the 0D flow model, monoenergetic neutrons are used. This is because the purpose of the irradiation is to generate group parameters for a given neutron irradiation energy rather than for a particular spectrum.

3.1.3 Moltres

Overview

MOOSE is an open-source, parallel finite element framework for which many applications have been built to handle different physics [50]. Coupling of these physics is simplified using MOOSE, and there are many pre-built applications available to handle a variety of physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they are used in Moltres [103, 82, 105, 102].

Moltres is an open-source MOOSE application which can be used for simulation of MSRs by solving the multi-group neutron diffusion and thermal-hydraulics equations [83]. This work uses Moltres to simulate the MSRE with different DNP group parameters. Moltres enables a validation study by comparing experimental results with different DNP group parameters. The validation study demonstrates the efficacy of these updated DNP group parameters compared to static group parameters.

The MSRE has experimental data which can be used for validation, the MSRE primarily consists of flow through channels which is simpler to model than open-pool designs, and the combination of Moltres and the MSRE has been previously demonstrated [83]. Since the initial demonstration, Park et al. have discussed the results, stating the following: "The neutron flux and temperature distributions agreed qualitatively with legacy MSRE data, albeit with some minor quantitative discrepancies due to simplifications and assumptions in the reactor geometry. Moltres has since undergone further development to support 1) the looping of DNP drift back into the reactor core, 2) coupling the aforementioned DNP drift to incompressible Navier–Stokes velocity flows within the reactor core, and 3) a decay heat model to simulate decay heat from fission products." [102]. Additionally, Park has released a dissertation with 1-D, 2-D, and 3-D neutronics models of the MSRE [101].

The purpose of using Moltres in this work is to simulate transients using MSR specific DNP group parameters and to generate those updated parameters in a higher-fidelity manner than the 0D method. In Moltres, there are three coupled physical phenomena required to model an MSR. These phenomena include the neutron flux, DNPs, and thermal-hydraulics.

Flux Modeling

The neutron flux is important to model, as it provides the energy of neutrons interacting with the sample as well as the number of interactions within the sample. This will directly influence the number of fissions which occur in the sample and the rate at which they occur, which directly affects the fission product concentrations, specifically the DNPs. The time-dependent multi-group neutron diffusion equation is given in Equation (2.27), recreated here:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\ & - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned}$$

In accounting for the neutron flux, consideration must also be given to the DNPs. The DNP contribute to the neutron flux and have their own energy spectra, so their contribution must be included.

Precursor Modeling

The time dependent DNP concentration is given in Equation (2.28), recreated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\nu \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i$$

An issue arises in using this equation however, as one of the primary motivations for using Moltres is to generate MSR specific DNP group parameters. However, this equation requires using DNP group parameters. To generate the DNP group parameters, this equation is instead replaced by (2.51), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

Specifically, the solution to this problem is to use spatially resolved depletion over the irradiation simulation. This means that instead of concentrations of DNP groups, the concentrations will be instead be of the DNPs themselves.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow using Equations (3.1) and (3.2) recreated from [101]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{ref}) \quad (3.1)$$

and

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where

ρ = fluid density

p = pressure

μ = dynamic viscosity

α = coefficient of thermal expansion

$\vec{g}$ = gravitational force vector

T = fluid temperature

T_{ref} = reference temperature at which the nominal density is provided.

The temperature distribution is given by Equation (3.3) recreated from [101], which shows the temperature advection-diffusion as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

and by Equation (3.4) recreated from [101], which gives the fission heat source term as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where

c_p = specific heat capacity of molten salt

k = effective thermal conductivity of molten salt

Q_f = fission heat source

ϵ_g = average fission energy released by neutrons in group g ,

Q_s = heat sink/removal.

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomena which must be considered to complete the model. This phenomena is related to the movement of the fuel salt boundary condition, as it is the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry, with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux is zero in the ex-core region and the eddy viscosity is zero, which leads to Equation (3.5), which shows the precursor concentration in the ex-core region is:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 Delayed Neutron Precursor Group Generation

To create DNP group parameter data in a reactor agnostic manner, the manner of generation needs to be reactor agnostic as well. Additionally, the group parameters must be generated for various parameters in such a way that the resulting group parameters are easily tabulated. However, a large number of parameters to fully detail the flow regime is not computationally feasible. Instead, the flow is reduced to only two parameters: an in-core average residence time (τ_{in}) and an ex-core residence time (τ_{ex}).

This approach captures the advection of the fuel salt around the primary loop as well as an approximation of turbulent and diffusive mixing. The terms are based on the averaged behavior from a control volume of fuel salt, so thermal-hydraulic effects such as vortices would be treated as changes in the residence times. This method allows

an approximate fitting for any circulating fuel reactor geometry, where all that is required is generating the relevant parameters for accessing the data.

The other parameters required are for the chemical removal, which can vary both in which fission products are removed and their reprocessing rates. Refueling is not considered in this work, as for the short time frames analyzed, the effects of adding additional fuel are considered to be negligible. Additionally, since the saturation irradiation time frame is on the order of multiple half lives of the longest lived DNP, this means a maximum irradiation time of five minutes is required to reach equilibrium DNPs concentrations. This means that elements which have cycle times on the order of months to years are likely to have negligible changes and do not need to be treated. Elements with changes on the order of seconds to days will have a sufficient contribution such that their inclusion is needed to generate accurate data.

The MSBR reprocessing scheme is used as the base for cycle times in this work, shown in Table 3.1. The rare earths, seminoble metals, volatile fluorides, discard, and higher nuclides are likely negligible for the time scales studied in this work. To ensure that they are negligible, the effects of these larger cycle time terms are studied in Section 4.1.1. Additionally, to better understand how the reprocessing rates affect the group parameters, the cycle times are scaled to various degrees in Section 4.1.2.

Table 3.1: MSBR Online Reprocessing Cycle Times from [115].

Reprocessing Group	Element(s)	Cycle Time (Full Power)
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	50 days
Rare Earths	Eu	500 days
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	20 seconds
Seminoble Metals	Zr, Cd, In, Sn	200 days
Gases	Kr, Xe	20 seconds
Volatile Fluorides	Br, I	60 days
Discard	Rb, Sr, Cs, Ba	3435 days
Salt Discard	Th, Li, Be, F	3435 days
Protactinium	Pa	3 days
Higher Nuclides	Np, Pu	16 years

Overall, the parameters which are used are incident neutron energy, target fissile nuclide, in-core residence time, ex-core residence time, mixing rate, reprocessing scheme, and reprocessing scaling vector. Because each unique parameter is an extra dimension of data that must be collected, it becomes increasingly expensive to compute all of the group parameters for each unique combination of data. In this work, the reprocessing scheme uses the MSBR reprocessing scheme and a single scaling scalar, rather than a vector containing scaling values for each element. The incident neutron energies used are 0.0253 *eV* and 1 *MeV*. For the residence times, a study is carried out in Section 4.1.2 to determine the minimum number of residence times and their values to capture general trends.

3.2.1 0D Flow Model

To generate sufficient data for each combination of input parameters, a simplified model is necessary. This simplified model will also provide estimates for the magnitude of variation which can be expected, as well as general trends. Specifically, this model treats the flow in a zero dimensional manner by using in-core and ex-core residence times. The fission history function is a square-wave, where the fission rate is some positive constant for the in-core residence time and zero for the ex-core residence time. This approximation simplifies the analytical solve required to create the non-linear least-squares problem, as shown in Appendix A.

Figure 3.1 shows how this fission history is used in OpenMC in order to mimic the irradiation on a control volume of fuel salt due to advection. The model takes in eight parameters to pass into OpenMC [118]: the in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. Of these terms, three, shown in green, are not varied to generate new group parameters: the temperature, target density, and total irradiation time. The other inputs are in yellow and vary to generate new group parameters. More detail on this block is provided in Section 3.3.

After OpenMC finishes the fixed source simulation, the final concentrations of nuclides in the irradiated sample are saved for analysis in custom scripts. These scripts take in the concentrations and combine them with the IAEA beta-delayed neutron emission database data to generate a delayed neutron count rate function [37]. A non-linear least squares fitting function from SciPy is then used to generate the half life and group abundances [141]. These group parameters are for a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. More detail on this block is provided in Section 3.4.

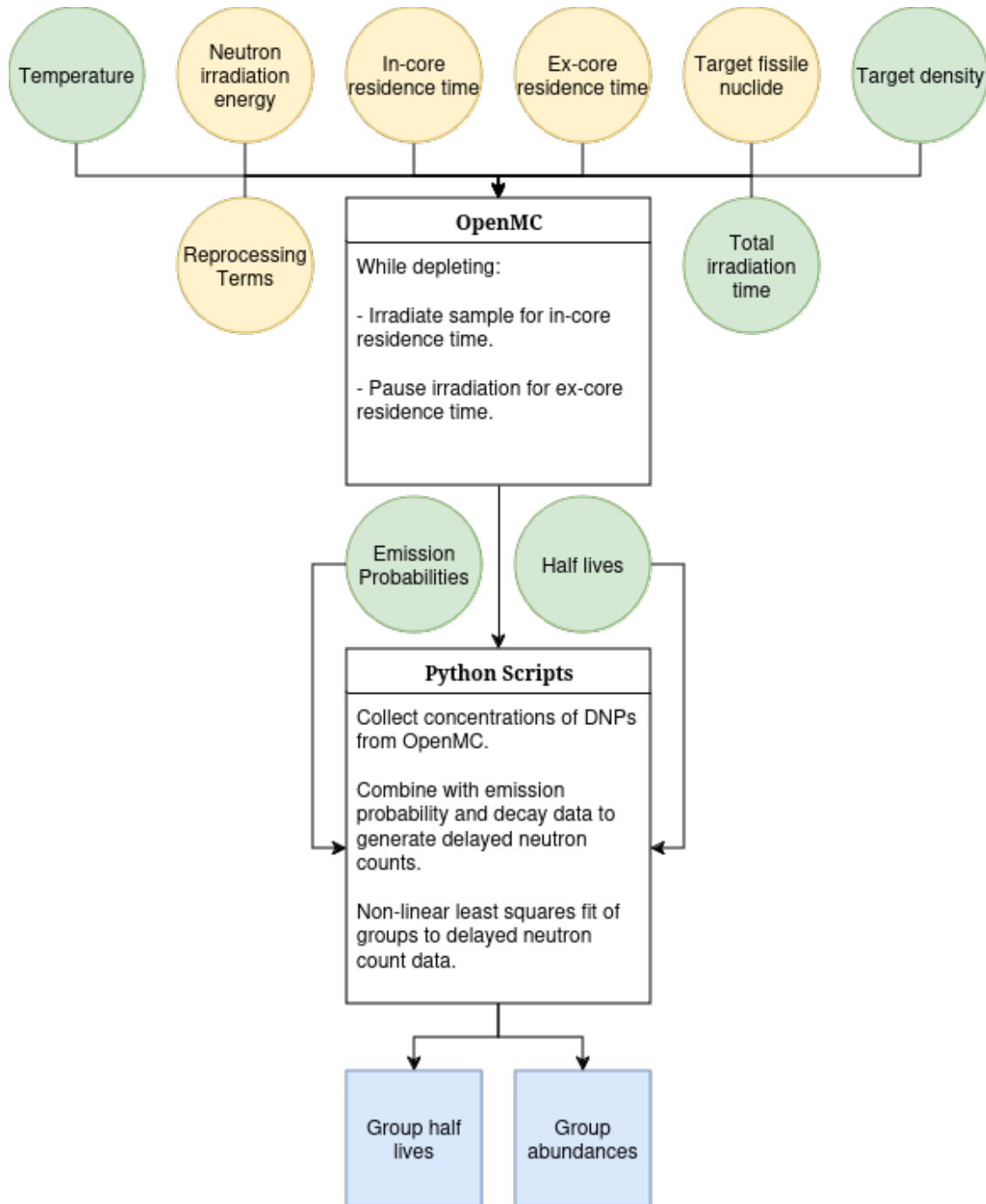


Figure 3.1: Flow diagram of 0D flow model used to generate flow-inclusive precursor group fit parameters.

There are several considerations required for the OpenMC simulations. The constant input parameters require some amount of consideration, which is discussed in Section 4.1. In particular, self-shielding of the sample due to density while avoiding self-multiplication, variations in cross sections due to temperature, and under or over irradiation can all affect the results and must be considered [66].

The general form of the OpenMC model is a three gram spherical sample with vacuum boundary conditions. This is treated as a fixed-source problem, with a uniform neutron source distributed evenly throughout the spherical

sample. This is not physical, as in reality the neutron source would have a spectrum and would be external. However, for the 0D flow model, the desired effects to capture are the differences between the group parameters with flow and reprocessing compared to the static results. This means such an approximation does not hinder the effectiveness of this 0D flow model.

Another consideration for the OpenMC model is having a consistent source particle rate. This is not strictly necessary for group parameter generation, as the values are per-fission, but it is convenient for data analysis, as the tallies are based on the source particles. The source rate for different simulations is given in Equation (3.6) as:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.6)$$

where

- $\dot{S}$ = the source particle rate [$\frac{n}{s}$]
- S_{net} = the total number of source particles [n]
- τ_{in} = the in-core residence time [s]
- τ_{ex} = the ex-core residence time [s]
- T = the net irradiation time [s].

3.2.2 Spatially Resolved Model

The 0D flow model offers a rapid generation of the DNP group parameters, though it treats the flow in a simplified manner. Rather than modeling the actual flow of the fuel salt, the residence times of the fuel salt in each region gives a rough estimate of the fuel circulation. Figure 3.2 shows the flow diagram for a spatially resolved model that uses Moltres instead of OpenMC.

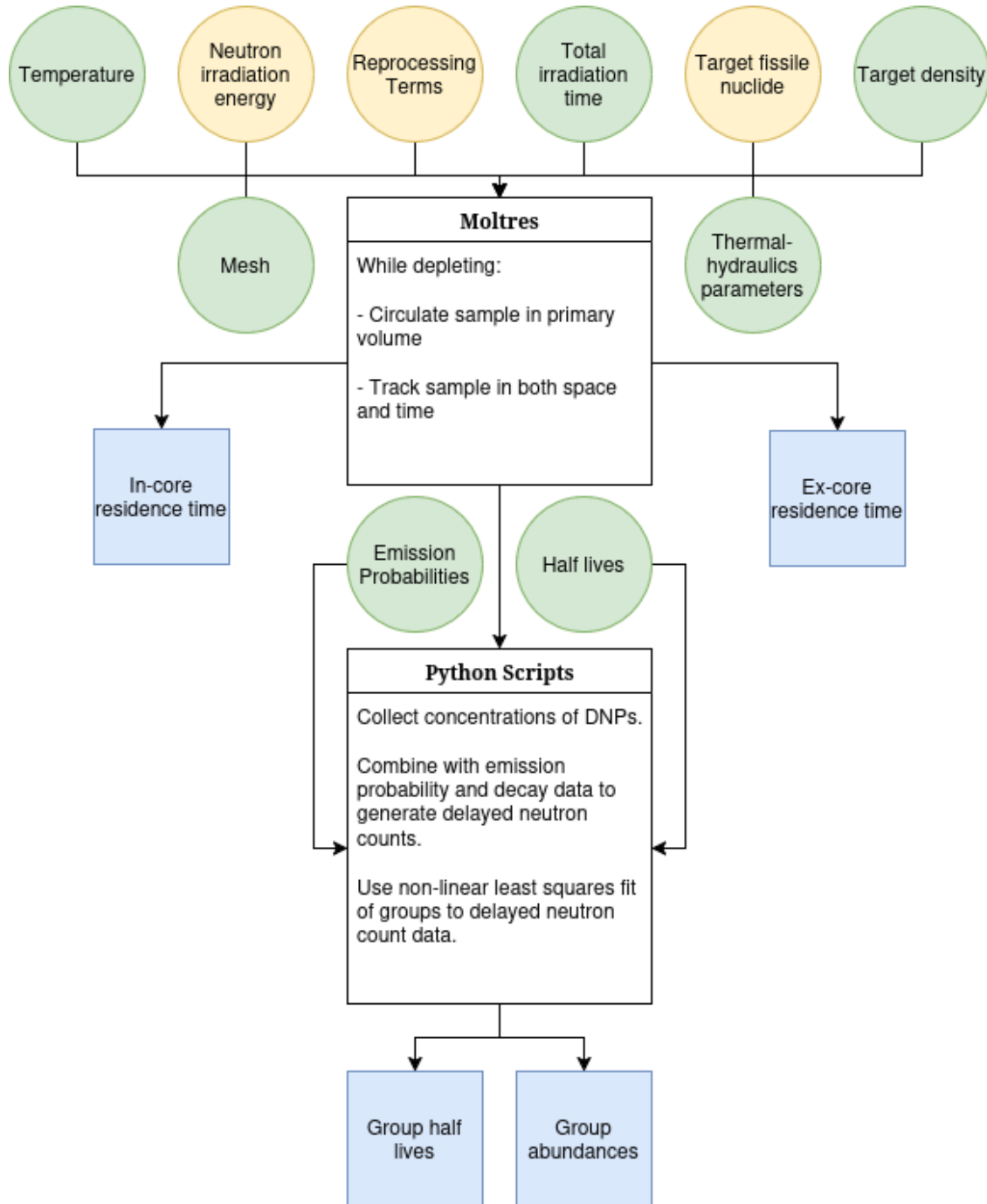


Figure 3.2: Flow diagram of spatially resolved flow model used to generate flow-inclusive precursor group fit parameters.

There are two new parameters in the spatially resolved flow model that are not present in the 0D flow model. The first is the mesh, which dictates the geometry of the problem. MOOSE offers built-in meshing tools, though this may be insufficient depending on the complexity of the problem geometry. In such a case, an external meshing tool, such as Gmsh, can be used [49]. The other new parameter is the thermal-hydraulics parameters. This parameter captures many of the physical phenomena involved with inclusion of spatial resolution, as discussed in Section

3.1.3.

The in-core and ex-core residence times are also outputs rather than inputs for the spatially resolved model. This is because the residence times can be calculated from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times will change over time as the irradiated sample diffuses and mixes more in the fuel salt. The net residence time, τ_{net} , will not change over time without some change to the model such as an increased temperature difference or induced flow rate. Equation (3.7) gives the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, t) dt d\vec{x}}{\int_V \int_0^t N(\vec{x}, t) dt d\vec{x}} \tau_{net}, \quad (3.7)$$

where

V_{in} = the in-core volume

V = the total primary volume

$N(\vec{x}, t)$ = the time and space resolved sample concentration

τ_{net} = the net residence, or circulation, time of the fuel salt.

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.8) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.8)$$

3.3 Delayed Neutron Count Rate Generation

Although the overview of the 0D flow model has already been discussed, details on the execution were not. In the following sub-sections, specific information is provided on the methodology behind the OpenMC block. This includes a deeper discussion on how the concentrations are generated from OpenMC, as well as the data which is used with those concentrations to generate a delayed neutron count rate profile.

3.3.1 Delayed Neutron Precursor Concentrations

The OpenMC model uses a fixed-source problem on a three gram spherical sample of a given fissile target. A uniformly distributed source is placed inside the target, and a vacuum boundary condition is used at the edge of the sphere. The sample must have a sufficient volume such that good statistics are achieved, but a volume that is too large can increase self-shielding and/or self-multiplication. This can be adjusted by modifying the density, where a more dense target increases self-shielding and self-multiplication. A less dense target reduces the probability of fission, which worsens the statistics for the solve.

The fixed-source used in this model is made up entirely of neutrons at a given energy, rather than a spectrum. This energy affects the fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or a range of energies. However, the macroscopic experiments, discussed in Section 2.1.1, are performed on a reactor which has a neutron energy spectrum, not mono-energetic neutrons. This could be worked around by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectra. However, for this work, the effects of the spectra are considered to be sufficiently small in

scale that they do not need to be included. There are also photons which can generate delayed neutrons through photofission, though these are not investigated in this work [68].

The distribution of the neutron source does not precisely describe the actual experimental process, where neutrons irradiate the sample from around the sample rather than inside. However, having the source evenly distributed removes concerns of self-shielding from the problem. The exact spatial locations where fission events occur within the sample should cause negligible changes in the concentrations of nuclides, which is the output from the OpenMC simulation. Assuming a density of $10 \frac{g}{cc}$, then the spherical sample has a radius of approximately 0.4 cm . The spatial resolution of the concentrations in such a small sample should not be significant. Additionally, the concentrations extracted from the OpenMC simulation are spatially collapsed in the post-processing, so the spatial effect of a uniformly distributed source serves only to reduce any self-shielding effects which could reduce the statistics and cause more transmutation than expected.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final time step of irradiation. This final time step of concentrations can then be passed along to the other scripts. These other scripts handle conversion of concentrations to the delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, the concentrations of each nuclide as a function of time for the measurement period can be determined. The concentrations are calculated as a function of half life, as shown in Equation (3.9) where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.9)$$

This is a simple equation which does not take into account decay chains between nuclides. An alternative approach which does account for these decay chains is to use repeated decay simulations of OpenMC in order to generate concentrations as a function of time. These concentrations can be used at coarse time steps and linearly interpolated using SciPy, enabling the scripts to access these more accurate concentrations at any time after irradiation of the sample [139]. This approach does not account for particles generated during decay, as it is considered beyond the scope of this work [26]. A comparison of these two approaches is investigated in Section 4.1.1.

The concentrations are combined with Equation (??), recreated here:

$$n_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t).$$

The efficiency term, ϵ , is set to 1 in this work, since the simulated method provides a count rate that includes every produced delayed neutron. Each DNP has its contribution over time summed up, generating the delayed neutron count rate.

3.4 Delayed Neutron Precursor Group Reconstruction

Once the delayed neutron count rate is generated, it can be converted into DNP group parameters. The process of fitting the group parameters requires a non-linear least squares method, as both the group half lives and group abundances must be determined. The equation describing the problem is given in Equation (2.3), recreated here:

$$n_d(t) \approx \epsilon F_s \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t}.$$

An important consideration for this process is the time values used when generating the group fits. An increased number of times means increased computational cost, but too few times results in a poorer fit [99]. This is investigated in Section 4.1.1.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation fit than it is for the saturation irradiation fit. This is due to the multiplication of the group yield by the group decay constant. Unfortunately, attempts to replace the multiplication of these two values with a new, independent term results in a poor fit. This is because the multiplication of the terms is not independent and is necessary to properly model the fit.

There is also linearization and normalization of the data in the non-linear least squares. The normalization is handled by taking the delayed neutron count rate, dividing by the fission term ¹, and then taking the natural logarithm. The normalized count rate is then passed into the non-linear least squares fitting. The linearization is handled by taking the natural logarithm of the delayed neutron count rate. This treatment reduces both the covariance and the χ^2 of the resulting non-linear least squares fit. Bounds are also set from zero to 1000 for each term to speed up the process, while the tolerances for convergence are set slightly above machine tolerance at 2.23×10^{16} to ensure the converged solution is well-refined. These convergence criteria include both change in the independent variables and the norm of the gradient, with termination occurring when either criteria meets the set tolerance.

The sample self-multiplication, mentioned in Section 2.1.1, is also a possible factor. However, since the number of fissions in the sample are taken directly from the OpenMC tallies, this does not become an issue which must be addressed.

Uncertainties are tracked in this work using the uncertainties package [41]. Uncertainties in the concentrations generated from OpenMC are not accounted for, but uncertainties in the experimentally measured emission probabilities and half lives are. These uncertainties are used in the minimization of χ^2 , as shown in Equation (2.6).

Once the group fits and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. The saturation irradiation provides better statistics for longer-lived groups, while the pulse irradiation is better for shorter-lived groups. Brady combines the irradiation results by using four long-period groups for saturation irradiation and four short-period groups for pulse irradiation. The groups are renormalized at the second group, where the two longest lived group half-lives and the ratio of yields use the saturation results [21]. The remainder of the groups use the pulse irradiation results.

For this work, a similar approach is used, though slightly simplified. The group parameters are calculated for pulse and saturation irradiation, after which the two longest lived groups use the saturation irradiation results while the four shortest lived groups use the pulse irradiation results.

¹The fission term is either the net number of fissions or the fission rate, depending on the irradiation type.

Chapter 4

Preliminary Results

For the preliminary results, all possible variations in the input parameters are not resolved. Instead, a default approach is utilized, with variations for refining the parameters. Thermal fission of ^{235}U is used with the MSBR reprocessing scheme with 10 second in-core and 10-second ex-core residence times at a temperature of 298.15 K and a density of 10 grams per centimeter cubed irradiated over seven minutes with linearly spaced time nodes and decay daughter tracking for the group parameter non-linear least squares fit.

4.1 Sensitivity Studies

4.1.1 Constants for Simulations

These sensitivity studies are used to analyze parameters that are held constant during the simulations and determine if simplifications can be made to some with negligible impact on the results. For these simulations, the parameters, unless otherwise specified, are given in Table 4.1.

OpenMC batches and OpenMC particles both refer to the settings in OpenMC which dictate the number of batches per generation and the number of particles per batch, respectively. The source particles parameter dictates the strength of the neutron source. The number of source particles is divided by the irradiation time to generate the neutron source rate used in OpenMC during the irradiation, as shown in Equation (3.6). The irradiation time is the amount of time the sample is irradiated, the decay time is the amount of time the sample has the delayed neutron count rate measured after irradiation, and the decay time step is the time step used when updating the nuclide concentrations post-irradiation. Additionally, by default, no chemical removal is modeled.

OpenMC Particles

If this value can be reduced, then the total amount of simulation time can be reduced. Alternatively, the default value may be insufficient for this problem and lead to inaccurate results. The model is exceedingly simple, composed of only a small sphere of fissile material and a vacuum boundary condition. Table 4.2 shows the total delayed neutron yields and average half-life for varying numbers of simulated particles in OpenMC. Because more particles means a slower simulation, a balance must be found between accuracy and cost. The differences between the largest number of particles for the total delayed neutron yield, $\bar{\nu}_d$, and the average half-life, $\bar{T}$, are given in the final two columns of the table. Based on these values, 50000 particles results in a fit accurate for the average half-life and total yield while also being somewhat computationally inexpensive.

Parameter	Value	Units
OpenMC Batches	10	-
OpenMC Particles	50000	-
OpenMC Depletion Chain	ENDF/B-VIII.0	-
OpenMC Cross Sections	ENDF/B-VIII.0	-
Source Particles	4.2×10^{16}	-
Temperature	920	K
Neutron Energy	0.0253	eV
Sample Nuclide	^{235}U	-
Sample Density	10	$\frac{g}{cm^3}$
In-core Residence Time	420	s
Ex-core Residence Time	-	s
Irradiation Time	420	s
Decay Time	420	s
Decay Time Step	1	s

Table 4.1: Default parameters used for sensitivity studies of constant parameters used in 0D flow DNP group parameter formulation OpenMC simulations

Number of Particles	$\bar{v}_d$	$\bar{T}$	$ \Delta \bar{v}_d $	$ \Delta \bar{T} $
1	0.016339	7.343276	0.002400	0.028600
10	0.019469	7.313167	0.000700	0.001500
100	0.018841	7.314828	0.000100	0.000200
500	0.018810	7.314185	0.000100	0.000500
1000	0.018693	7.315162	0.000000	0.000500
5000	0.018667	7.314440	0.000100	0.000200
10000	0.018700	7.314947	0.000000	0.000300
50000	0.018741	7.314586	0.000000	0.000100
100000	0.018730	7.314608	0.000000	0.000100
500000	0.018731	7.314636	0.000000	0.000000
1000000	0.018733	7.314663	-	-

Table 4.2: Sensitivity study of total delayed neutron yields and average half-lives due to number of particles simulated in OpenMC.

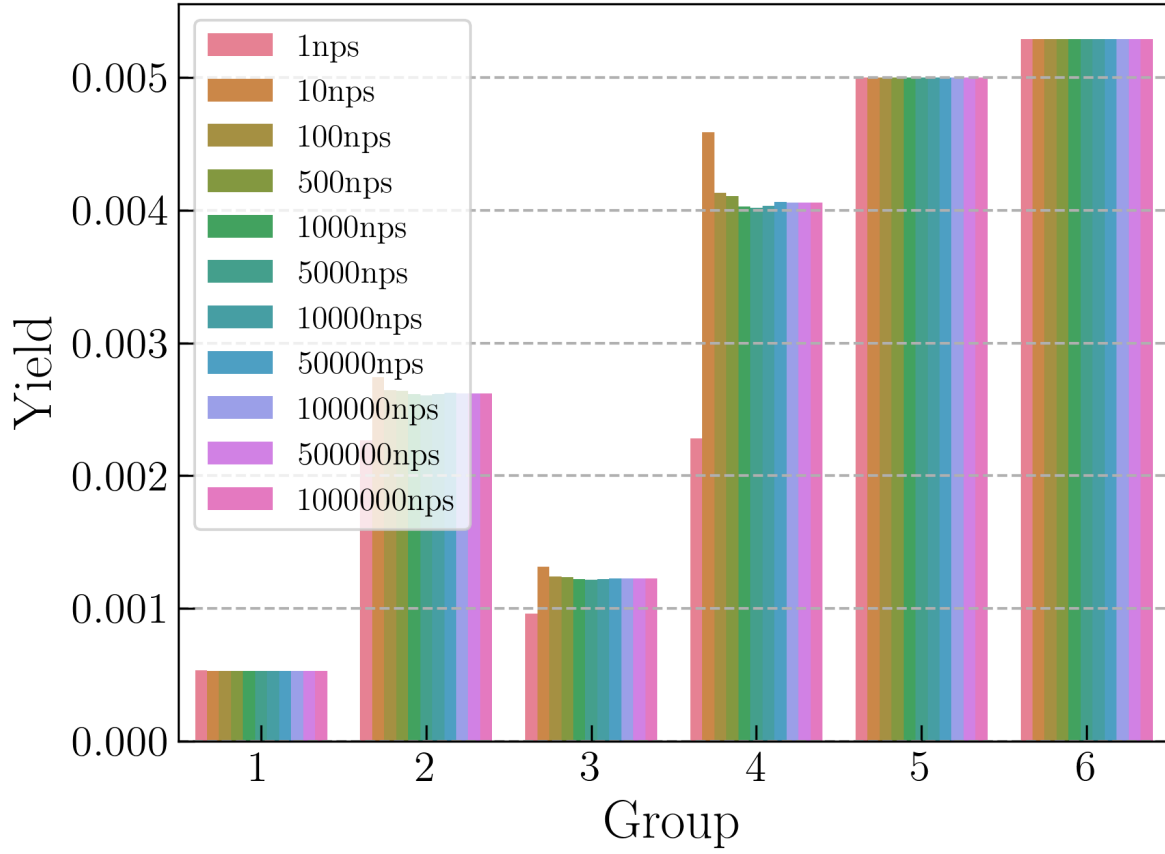


Figure 4.1: Group yields resulting from various numbers of simulated particles in OpenMC.

The variation in these values for lower particle counts is driven by differences in the group 2, 3, and 4 fits. Figure 4.1 shows the group yields for each different number of particles, where the largest differences are in groups 2, 3, and 4. These group half-lives range from 4.4 to 24 seconds. The cause of this difference must be from differences in nuclide concentrations post-irradiation, as the emission probability and decay constant data are the same between the different simulations compared here. To evaluate which nuclides are causing this difference, the standard deviation of each set of concentrations is divided by the mean, which results in the coefficient of variation (CV) for each nuclide. Some of the nuclides with the largest CV have a negligible contribution to the delayed neutron yield, such as ^{209}Tl . To scale by this consideration, the CV is multiplied by the total number of delayed neutrons emitted from that nuclide i , $\bar{\nu}_{d,i}$, the equation for which is given in Equation (2.21), replicated here:

$$\bar{\nu}_{d,i} = P_{n,i} \lambda_i N_{i,0}$$

The scaling of the CV by the total delayed neutron yield is important because nuclides like ^{209}Tl have a CV of 9.65%, but a total delayed neutron yield of 0. The nuclides are sorted by a new scoring term, α , where the full term in Equation (4.1) is given as:

$$\alpha_i = \frac{\bar{\nu}_{d,i} CV}{\bar{\nu}_d} \quad (4.1)$$

Based on α , only two of the top ten nuclides are within groups two through four, with the nuclides being ^{89}Br and ^{137}I . When sorting by CV instead, none of the top thirty nuclides are within groups two through four. If the CV is replaced with the standard deviation, then there are six out of ten nuclides which are within groups two through four. These nuclides are ^{137}I , ^{88}Br , ^{93}Rb , ^{89}Br , ^{138}I , and ^{136}Te . Although the relative changes in these nuclides are smaller, the absolute change is larger, which drives a greater difference in groups 2 through 4.

Temperature

Target Density

Irradiation Time

Decay Time and Decay Time Steps

The bulk of the computational cost comes from writing statepoint files; sufficient resolution is required to ensure a smooth decay curve.

Large Cycle Time Elements

Decay Daughter Tracking

4.1.2 Variables for Simulations

These sensitivity studies are used to analyze the parameters that change between simulations to understand the effect these changing parameters have on the results.

Removal Rates

In and Ex-core Residence Times

Combined Removal Rates and Residence Times

4.2 Verification and Validation

4.2.1 0D Flow Model Verification

[81] table 2 has group parameters, these can be used as a direct comparison and the effects can be compared for the MSFR.

[90] has MSFR PRK and IPK which can be compared.

fluid flow velocity change on power (Dulla PhD thesis I can't access)

[76] - can compare reactivity insertion effects on power

[90] - extracted power effect on power

[87] -

Microscopic group parameters [77, 93, 62] should match static results (no flow or chemical removal.)

ENDF Table II in [62] shows a yield of 1.9 for thermal u235 in ENDF, which matches the static results in the simplified model simulation.

4.2.2 0D Flow Model Validation

- MSRE pump start-up and coast-down with Moltres and Quasimolto [109, 111].
- Replicate Godiva irradiation or similar to match experimental results

Chapter 5

Proposed Work

5.1 Develop Sophisticated Model

5.1.1 Group Generation

- Moltres can be used to perform depletion (Nathan Glaser's work)
- Can have spatially resolved irradiation using this functionality (while accounting for thermal-hydraulics)
- Flow analyses can be conducted (channel flow, open core, different fuel diffusion coefficients)
- Verify if 0D flow model can be used to generate group parameters (if so, when; else why not) (reprocessing from ex-core vs OpenMC dimensionless reprocessing)
- Compare computational cost to accuracy

5.1.2 Group Evaluation

- Moltres takes in group parameters to evaluate reactor transients
- Various safety and safeguards scenarios can be investigated
- Scenarios where the difference between static and flowing group fits leads to incorrect modeling can be discussed
- Comparison with 0D parameters

5.1.3 Least-Squares Fission History

See Appendix A. This will probably need numerical evaluation of integral with a polynomial fit afterwards. Could also include $R(t)$ using $N(t)$ rather than assuming exponential decay.

5.2 Polynomial Fits of Tabulated Parameters

Include polynomial fits (figures) showing how group parameters vary based on tabulated parameters (in-core, ex-core, chemical removal scaling, etc.)

Comparison with 0D Flow Model

5.3 Generate Delayed Neutron Precursor Group Spectra

- [33] discusses why it's important in section 6
- Discuss how spectra could be resolved
- Can also be compared between higher fidelity model

5.4 Uncertainty Quantification

Kozlowski paper Using sensitivity to flow and reprocessing. Uncertainty in half lives and emission probabilities is small. Neglect uncertainty in concentrations from OpenMC (cross sections, fission yields, half lives, transmutations, decay yields, etc.).

5.5 Other extensions

- Model more nuclides to see if analyses are valid for all nuclides or only some (create polynomial fits to describe behavior)
- Look into different ways of combining pulse and saturation irradiation results [92]
- Diffusion effect (not really modelable in time pulses only unless multiple materials are used) (could be approximated in 0D Flow Model).
- Photoneutrons - investigate
- Spatially resolved flux in 0D Flow Model - consider, Specific neutron spectra
- Thermochemical effects - different nuclides will have different chemical removal rates, which will affect results (Sam knows more about this and could help me figure out if there would be something here in particular worth investigating)
- Evaluation of other datasets impact on results [62, 87]

Appendix A

Appendix A: Derivation Delayed Neutron Count Rate in 0D Flow for Equal Residence Times

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [143]:

$$R(t) = \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and,

$$\dot{n}_d(T, t) = \bar{v}_d \int_0^T F(\tau) R(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where,

T = the length of the fission history [s]

t = the decay time from the end of the fission history [s]

$F(\tau)$ = the fission rate history [$\frac{\text{fissions}}{\text{s}}$].

Because the results are normalized to a single delayed neutron, the $\bar{v}_d$ term is divided out.

To derive an analytical solution to this problem for the 0D flow approach where the in-core and ex-core residence times, τ_{in} and τ_{ex} , are equivalent, an approximation of the fission history must be made. For this derivation, τ_{res} will be used to represent both the in-core and ex-core residence times. Because the 0D flow model's fission history is a positive square wave, analytical integration is not feasible. Instead, a Fourier decomposition of the square wave into a sum of sine waves makes the problem far more approachable. This is given as:

$$\frac{F(t)}{F_0} = \frac{1}{2} + \frac{2}{\pi} \left(\sin\left(\frac{\pi t}{\tau_{res}}\right) + \frac{1}{3} \sin\left(\frac{3\pi t}{\tau_{res}}\right) + \dots \right), \quad (\text{AA.3})$$

where,

F_0 = the magnitude of the constant fission rate [$\frac{\text{fissions}}{\text{s}}$].

Plugging this in and separating the integral, for group k and for j Fourier terms, gives:

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} \lambda_k}{2} \int_0^T e^{-\lambda_k(T-\tau+t)} d\tau \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k}{\pi} \int_0^T \sin\left(\frac{\pi\tau}{\tau_{res}}\right) e^{-\lambda_k(T-\tau+t)} d\tau \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k}{(2j+1)\pi} \int_0^T \sin\left(\frac{(2j+1)\pi\tau}{\tau_{res}}\right) e^{-\lambda_k(T-\tau+t)} d\tau
\end{aligned} \tag{AA.4}$$

Solving the integrals and simplifying yields:

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T})}{2} \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k^2 \tau_{res}^2 e^{-\lambda_k t}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \sin\left(\frac{\pi(2j+1)T}{\tau_{res}}\right) \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \cos\left(\frac{\pi(2j+1)T}{\tau_{res}}\right) \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} 2\pi j \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \pi,
\end{aligned} \tag{AA.5}$$

where each integrated Fourier term generates four terms added to the sum.

Similar to [143], the irradiation time is taken to the limit as it approaches infinity, though only for the non-periodic terms. By irradiating for several half-lives of time for each DNP, where the longest half-life is roughly one minute, then the irradiation time can be taken as infinite. There are two reasons this limit is taken. Firstly, a pulse irradiation with circulation will not provide any unique results when compared to traditional data which already exists. Additionally, this allows the equation to be simplified, as shown in Equation (AA.6). Only the sine term does not go to zero as the irradiation time goes to infinity, so it is the only term significant for saturation irradiation in the 0D flow model.

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} e^{-\lambda_k t}}{2} \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k^2 \tau_{res}^2 e^{-\lambda_k t}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \sin\left(\frac{\pi(2j+1)T}{\tau_{res}}\right)
\end{aligned} \tag{AA.6}$$

The final assumption to make is that the irradiation time, T , is an integer multiple of the residence times, τ_{res} . This assumption makes the sine term 0 for every value of k , which means that the result simplifies to:

$$n_d(t) = \frac{F_0}{2} \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \tag{AA.7}$$

This is very similar to the saturation irradiation delayed neutron count rate equation, which is:

$$n_d(t) = F_0 \sum_{k=1}^K \bar{\nu}_{d,k} e^{-\lambda_k t} \quad (\text{AA.8})$$

Because of the initial assumption made, which was that half the time is in-core and half ex-core, the average fission rate is half of the fission rate scaling term, F_0 . For both cases (constant irradiation and half in-core and half ex-core), the fission term in front is simply the average fission rate over the irradiation time.

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